

Quantification of Metabolic Activity by Converting Raw PET Data into SUV and SUVr Maps : Application to Alzheimer's Disease (May 2026)

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ABSTRACT

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by cognitive decline and alterations in cerebral metabolism. [18F]FDG positron emission tomography (PET) allows in vivo assessment of this metabolic dysfunction. However, the clinical exploitation of these images requires a rigorous quantitative approach. This work presents a methodology for converting raw PET data into Standardized Uptake Value (SUV) maps and Standardized Uptake Value ratios (SUVr). Applied to a cohort of healthy controls, patients with mild cognitive impairment (MCI), and patients with Alzheimer's disease (AD), this approach aims to characterize the metabolic profiles associated with each stage of the pathology and to discuss the limitations related to the heterogeneity of acquisition modalities.

MOTS CLÉS Alzheimer's disease, positron emission tomography, quantification, SUV, SUVr

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I. INTRODUCTION

Alzheimer's disease (AD) is the most common cause of dementia in elderly individuals. It is characterized by a progressive impairment of cognitive functions, particularly episodic memory, language, and executive functions. From a pathophysiological standpoint, it is based on two characteristic lesions : amyloid plaques (extracellular accumulation of β -amyloid peptides) and neurofibrillary tangles (intracellular aggregation of hyperphosphorylated tau protein). These processes lead to synaptic and neuronal loss, beginning in the hippocampus before spreading to associative cortical regions

[1][2].

Conventional diagnostic methods, based on clinical evaluation and structural imaging (MRI), show limitations for early detection. MRI only visualizes brain atrophy, which is a late marker. In contrast, functional alterations precede structural abnormalities. In this context, positron emission tomography (PET) using [18F]FDG stands out as a key tool for the early evaluation of cerebral metabolic dysfunction[3].

A. POSITRON EMISSION TOMOGRAPHY

Positron Emission Tomography (PET) using [18F] Fluorodeoxyglucose ([18F]FDG) is recognized as the "Gold Standard" for the in vivo assessment of neurodegeneration. While visible atrophy on MRI is a late structural marker, glucose hypometabolism detected via PET is an early indicator of synaptic dysfunction[4].

[18F]FDG, a glucose analogue, acts as a tracer for neuronal activity : it is taken up by GLUT transporters and trapped inside cells following phosphorylation. Its half-life of 109.77 min offers an optimal compromise between clinical logistics and image quality. According to the pivotal study by Mosconi et al., FDG-PET exhibits a sensitivity greater than 90% in identifying Alzheimer's Disease (AD) and predicting conversion from mild cognitive impairment (MCI) stages. In this context, quantification via SUV (Standardized Uptake Value) becomes an indispensable tool to objectively assess the severity of metabolic impairment compared to control subjects (CN)[5].

Although valuable, this measurement of brain metabolism requires a rigorous quantitative approach to be fully utilized. With this in mind, we detail below the quantification methods applied to PET images.

B. PET IMAGE QUANTIFICATION

In positron emission tomography, we aim to measure the concentration of a radioactive tracer within tissues, expressed in Bq/mL. To achieve this, it is possible to perform a qualitative, visual analysis. The latter relies on the expertise of the clinician and can be subject to a lack of sensitivity and rigor, particularly for detecting the early stages of Alzheimer's disease. This is why a quantitative approach is preferred. It enables the extraction of objective, reproducible numerical values and, by extension, facilitates comparing a patient against a database. In our study, two main methods are used, namely SUV (Standardized Uptake Value) and SUVr (Standardized Uptake Value ratio)[6] [7].

SUV is a quantification index generally used in clinical applications. It normalizes the activity measured in the image based on patient-specific parameters and is defined by the

expression :

$$SUV = \frac{\text{Radioactive concentration}_{\text{Region of interest}}(\text{Bq/mL})}{\frac{\text{Injected dose}(\text{Bq})}{\text{Patient weight}(\text{g})}}$$

Despite this, SUV remains sensitive to several factors, such as the delay between injection and acquisition, the patient's blood glucose level when FDG is used as a tracer, and even potential calibration errors of the equipment used. Thus, for neurological studies, this calculation alone is not sufficient, as a more stable reference is required[8].

This is where the SUVr (Standardized Uptake Value ratio) comes into play. It consists of dividing the activity of one or more target regions by that of a healthy reference region unaffected by the pathology.

$$SUVr = \frac{\text{Activity in Target Region}}{\text{Activity in reference area}}$$

In our study applied to Alzheimer's disease, it highlights abnormalities such as decreased glucose consumption and abnormal accumulation of amyloid and tau proteins, facilitating the differentiation between subjects with normal cognitive abilities (Cognitive Normal - CN), those exhibiting mild cognitive impairment (Mild Cognitive Impairment - MCI), and those who have reached the Alzheimer's disease stage (Alzheimer's Disease - AD).

In short, quantification turns a visual approach into a precise calculation of brain metabolism. It promotes the early detection of neurodegenerative processes, the objective classification of patients according to the stage and progression of their disease, and ensures the quality and relevance of clinical findings.

II. MATERIALS AND METHODS

A. STUDY POPULATION AND PET ACQUISITIONS

A total of 24 subjects were included in this study. Their distribution by clinical group, age, PET acquisition system, and measurement units is detailed in Table 1.

There are 10 patients with Alzheimer's disease (AD), 6 patients with mild cognitive impairment (MCI), and 8 cognitive normal control subjects (CN). Including healthy control subjects (CN) and patients with mild cognitive impairment (MCI) provides a complete clinical spectrum : from subjects without impairment to proven Alzheimer's patients. MCI represents a transitional state between normal aging and dementia. Their inclusion is essential to evaluate the early performance of PET biomarkers and to study metabolic progression toward Alzheimer's disease.

Note : Among these 24 subjects, one MCI patient and one AD patient were each included twice, in two different years. In our study, we treat them as independent subjects, as each PET examination corresponds to a distinct clinical and

metabolic state over time.

Positron emission tomography (PET) images using [18F]FDG were acquired according to a standardized protocol and performed across several systems : Philips GEMINI TF TOF 18, Siemens 1094 (majority in the AD group), Siemens Biograph TruePoint, GE Discovery ST, and an older CPS system (model 1023).

TABLE 1. Demographic, clinical, and technical characteristics of the included subjects

Patient ID	Age	Group	PET Acquisition	Units
5018	73	AD	Philips GEMINI TF TOF 18	CNTS
1059	85	AD	CPS 1023	Bq/mL
1257	85	AD	CPS_1023	Bq/mL
4136_11	66	AD	SIEMENS 1094	Bq/mL
4136_13	68	AD	SIEMENS 1094	Bq/mL
4152	60	AD	SIEMENS 1094	Bq/mL
4373	71	AD	SIEMENS 1094	Bq/mL
5165	78	AD	SIEMENS 1094	Bq/mL
6264	55	AD	SIEMENS Biograph TruePoint	Bq/mL
6854	86	AD	GE MEDICAL SYSTEMS Discovery ST	Bq/mL
0729	69	MCI	Philips GEMINI TF TOF 18	CNTS
1155_11	62	MCI	Philips GEMINI TF TOF 18	CNTS
1155_13	64	MCI	Philips GEMINI TF TOF 18	CNTS
1268	87	MCI	Philips GEMINI TF TOF 18	CNTS
4799	73	MCI	Philips GEMINI TF TOF 18	CNTS
0908	67	MCI	SIEMENS 1094	Bq/mL
0295	90	CN	Philips GEMINI TF TOF 18	CNTS
0413	81	CN	Philips GEMINI TF TOF 18	CNTS
0685	93	CN	Philips GEMINI TF TOF 18	CNTS
1261	77	CN	Philips GEMINI TF TOF 18	CNTS
1280	75	CN	Philips GEMINI TF TOF 18	CNTS
4213	78	CN	Philips GEMINI TF TOF 18	CNTS
4225	72	CN	Philips GEMINI TF TOF 18	CNTS
4262	74	CN	Philips GEMINI TF TOF 18	CNTS

An important methodological point must be highlighted : only acquisitions calibrated in Bq/mL allow for the calculation of SUV (*Standardized Uptake Value*) maps. However, as shown in Table 1, a large majority of examinations (including all CN subjects, 5 MCI subjects, and 1 AD patient) are expressed in CNTS (raw counts). Converting CNTS to Bq/mL requires a calibration factor specific to each system. It was not possible for us to obtain these factors. Consequently, only the 9 AD patients and 1 MCI patient whose acquisitions are in Bq/mL were retained for the SUV map generation step. The other subjects were excluded from this quantitative analysis.

B. CALCULATION OF THE SUV CONVERSION FACTOR (P_F)

Semi-quantitative measurement was based on calculating the Standardized Uptake Value (SUV). A conversion factor, denoted as p_f , was calculated for each patient to transform the PET images into SUV units.

From the DICOM metadata of each exam, the following parameters were extracted using a MATLAB script : patient weight (kg), total injected dose at time zero A_0 (Bq), injection start time T_0 , and acquisition start time T_a .

The half-life constant of [18F]FDG was set to $T_{1/2} = 109.77$ min. The residual activity at the time of acquisition A_a was calculated using the exponential radioactive decay law :

$$A_a = A_0 \times \exp\left(-\frac{\ln(2) \cdot \Delta T}{T_{1/2}}\right)$$

where $\Delta T = T_a - T_0$ (in min) is the time delay between injection and the start of acquisition. The factor p_f was then obtained by :

$$p_f = \frac{\text{weight (kg)} \times 1000}{A_a \text{ (Bq)}}$$

This factor allows converting a PET image expressed in Bq/mL into an SUV image. Taking radioactive decay into account is crucial : without this correction, the SUV would be systematically underestimated for patients who waited longer before acquisition. Normalizing by A_a ensures strict comparability between subjects[9].

C. IMAGE PREPROCESSING IN SPM

All processing steps were performed in MATLAB using the SPM25 toolbox. For each patient, raw DICOM images were imported in NifTI format via the DICOM Import interface. When multiple files were generated, only the last one (reconstructed volume with the highest quality) was kept.

The selected NifTI files (intensities in Bq/mL) were loaded into the ImCalc interface. Each voxel was multiplied by the factor p_f according to :

$$SUV_{\text{voxel}} = I_{\text{Bq/mL}} \times p_f$$

Individual SUV maps were then spatially normalized using the `batch` tool, removing inter-individual anatomical variability. Resampling via trilinear interpolation was applied to estimate values for new voxels with high precision while avoiding abrupt discontinuities. Isotropic spatial smoothing was then performed with a Gaussian kernel of full width at half maximum (FWHM) of 8x8x8mm. This smoothing improves the signal-to-noise ratio and aligns residuals with the assumptions of parametric statistical models. The resulting image, labeled `swSUV_<patient>.nii`, served as the input for all subsequent analyses.

For images intended for SUVr calculation, a 4D to 3D conversion was first performed, generating six *frames*. The sixth frame (acquired between 30 and 60 minutes post-injection) was selected because it corresponds to the plateau phase of FDG stabilization in brain tissue. Manual coregistration was then carried out using SPM's `Display` command : the origin of the PET image was reoriented to match that of the MRI template included in the software by adjusting translation and rotation parameters. Visual verification was performed using the `Check Reg` tool.

Spatial normalization was applied following the same principles as for SUV images. After spatial normalization, an anatomical atlas in MNI space was used to identify regions of interest. Each voxel contains a unique identifier associated with a specific anatomical structure (Volume of

Interest, VOI). Thanks to spatial normalization, voxel-to-voxel correspondence between the PET image and the atlas is seamless.

The cerebellum was chosen as the reference region. This area, largely spared by Alzheimer's disease, consumes glucose consistently and stably despite neurodegenerative processes. By dividing by the mean value of the cerebellum, individual variations (weight, injected dose) are removed to retain purely the metabolic contrast.

D. CALCULATION OF SUVR

The SUVR calculation was carried out in two steps.

First, a MATLAB script extracted voxels associated with the cerebellum from the atlas and computed the arithmetic mean of their intensities :

$$\text{Mean}_{\text{cerebellum}} = \frac{1}{N} \sum_{i=1}^N S_i$$

where N is the number of voxels in the cerebellum volume and S_i is the signal of each voxel.

Second, using SPM's `ImCalc` tool, each patient's normalized and smoothed PET image was divided by this reference value :

$$\text{SUVR}(x, y, z) = \frac{\text{Image}_{\text{PET}}(x, y, z)}{\text{Mean}_{\text{cerebellum}}}$$

Each voxel of the resulting SUVR image indicates how many times more or less active it is compared to the cerebellum. A voxel with a value of 1.2 consumes 20% more glucose than the cerebellum; a voxel with a value of 0.7 consumes 30% less.

E. VALUE EXTRACTION AND VISUALIZATION

Extraction of global SUV values for each subject was performed using the MATLAB script `rex.m` (ROI Extraction, based on SPM). The mask used was a gray matter mask in MNI space, named `MINI_grey_Mask.nii`. For each patient, the image `swSUV_<patient>.nii` was set as the source image, with the mask as the region of interest. The mean SUV value in the gray matter was extracted from the file `MINI_grey_Mask.rex.txt[10]`. The choice of gray matter is justified by the fact that it encompasses the brain regions richest in synaptic and neuronal activity, which are the main areas affected by the metabolic alterations observed in Alzheimer's disease. This approach thus yields a global, reproducible, and representative indicator of cerebral metabolism.

For each patient, normalized and smoothed SUV maps (`swSUV_*.nii`) were visualized in MATLAB along three orthogonal planes : axial, coronal, and sagittal. Slices were extracted at the midpoint of each image dimension using `spm_read_vols`. Visualizations were created with the jet color map and a scale bar. The purpose was visual quality control of the SUV maps. This step verified the quality of

spatial normalization and smoothing. Visual comparison of the maps also enabled a qualitative assessment of cerebral glucose metabolism distribution and highlighted characteristic regions of hypometabolism in Alzheimer's disease.

In addition to this global measure, regional SUV values were also studied to evaluate the spatial distribution of cerebral hypometabolism. This regional analysis makes it possible to identify the structures most affected by neurodegeneration and to compare metabolic profiles across different patients.

III. RESULTS

A. GLOBAL SUV

Global SUV values were extracted for 9 patients with Alzheimer's disease (AD) and 1 patient with mild cognitive impairment (MCI). The results are presented in Table 2.

TABLE 2. Global SUV by patient

Patient ID	Age	Dementia	Global SUV
0908	67	MCI	11.80
1059	85	AD	5.55
1257	85	AD	2.89
4136_11	66	AD	5.35
4136_13	68	AD	3.96
4152	60	AD	7.85
4373	71	AD	5.10
5165	78	AD	6.05
6264	55	AD	8.03
6854	86	AD	4.88

For the 9 AD patients, the mean global SUV is 5.52 ± 1.66 (median : 5.35). The extreme values range from 2.89 (patient 1257, 85 years old) to 8.03 (patient 6264, 55 years old). Figure 1 presents the distribution of SUVs as a box plot.

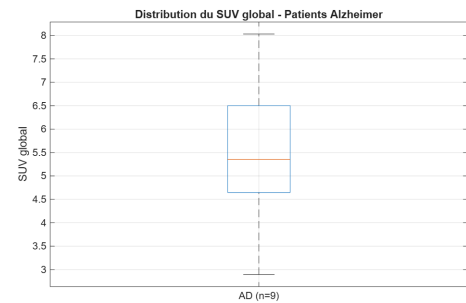


FIGURE 1. Distribution of global SUVs in the 9 AD patients.

A correlation analysis was performed between the age of the AD patients and their global SUV. Pearson's correlation coefficient is $r = 0.068$ ($p = 0.862$), indicating the absence of a significant linear relationship between these two variables. Contrary to the preliminary observation suggesting a trend toward decreasing SUV with age, the statistical analysis does not confirm this hypothesis. This lack of correlation can be explained by the limited size of our sample ($n = 9$) and by the inter-individual heterogeneity characteristic of

Alzheimer's disease.

The MCI patient (0908) stands out clearly with a global SUV of 11.80, approximately twice as high as the AD group mean (5.52). This elevated value could reflect a preservation of cerebral glucose metabolism at the early stage of the pathology, or even a compensatory hypermetabolism phenomenon, although the absence of other MCI subjects in our cohort prevents generalizing this observation. We will detail this feature with the SUVr analysis.

B. COMPARISON OF SUV MAPS : MCI VS. AD

To illustrate the metabolic differences between stages of the pathology, the SUV maps of the MCI patient (0908, global SUV = 11.80) and the AD patient (1257, global SUV = 2.89) were visually compared. Figure 2 presents the three orthogonal planes (axial, coronal, sagittal) for each of these two cases.

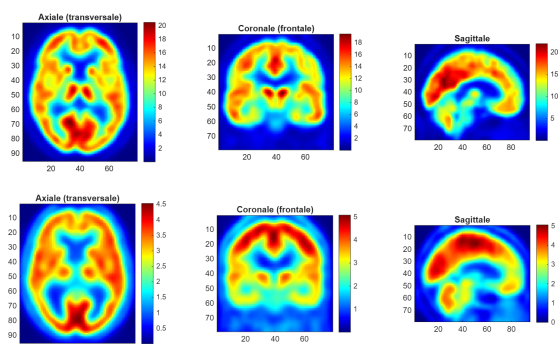


FIGURE 2. Comparison of SUV maps. (A) MCI Patient 0908, global SUV = 11.80. (B) AD Patient 1257, global SUV = 2.89.

A qualitative comparison was conducted between the SUV maps of the MCI patient (0908, global SUV = 11.80) and the AD patient (1257, global SUV = 2.89). It is important to note that the color scales (colorbars) differ between the two subjects, as they are calibrated to the range of values of each exam.

In MCI patient 0908 (Figure 2, top), the scale extends from 0 to 20. An intense and widespread uptake of [18F]FDG is observed, with maximum values reaching 20 in cortical regions, demonstrating preserved or even elevated glucose metabolism.

In AD patient 1257 (Figure 2, bottom), the scale extends only from 0 to 5 (axial), 0 to 5 (coronal), and 0 to 5 (sagittal). Overall activity is markedly decreased, with maximum values peaking at 5 in the most active regions. The visual appearance of a "more extended distribution" of activity in this AD patient is misleading : it is explained by severe brain atrophy, which reduces cerebral parenchyma volume and creates the illusion of a larger proportion of colored voxels,

whereas the absolute signal intensity is very low[11].

Thus, although both images use the same color palette (jet, from blue to red), the scale bounds are radically different. This difference quantitatively illustrates the severity of hypometabolism in the AD patient (maximum values at 5) compared to the MCI patient (maximum values at 20). However, these observations remain descriptive and cannot replace a rigorous quantitative analysis on a larger cohort.

C. REGIONAL SUVs AND CORRELATION

SUV values were also extracted for five regions of interest (frontal, parietal, temporal, occipital, and cerebellum) across the 9 AD patients. Table 3 summarizes these data.

TABLE 3. Regional SUVs in AD patients

Patient ID	Frontal	Parietal	Temporal	Occipital	Cerebellum	Global SUV
6854	4.79	4.73	4.39	5.47	5.69	4.88
5165	6.24	6.01	5.56	6.54	5.94	6.05
4373	4.90	4.94	4.63	6.02	5.76	5.10
4152	7.34	7.36	7.21	9.99	8.37	7.85
4136_11	5.58	4.98	4.58	5.66	6.12	5.35
4136_13	4.09	3.69	3.45	4.10	4.75	3.96
1257	3.00	3.01	2.56	2.93	2.93	2.89
1059	5.55	5.59	4.86	5.99	6.22	5.55
6264	8.00	7.64	7.55	8.96	7.87	8.03

In general, the occipital region shows the highest SUV values in the majority of patients (up to 9.99 for patient 4152), while the temporal region shows the lowest values (2.56 for patient 1257). This topography is consistent with the known pathophysiology of Alzheimer's disease, which relatively spares the occipital (visual) cortex while affecting temporal regions early, most severely for patient 1257 (85 years old). Furthermore, good concordance is observed between the global SUV calculated by SPM and the independently calculated global SUV (mean of the five regions), validating the consistency of our measurements (mean difference less than 0.1)[12].

In order to explore more precisely the influence of age on regional brain metabolism, a Pearson correlation analysis was performed between the age of the AD patients and the SUV values in five regions of interest. The results are presented in Table 4 and illustrated in Figure 3.

TABLE 4. Correlations between age and regional SUVs (AD group, n=9), NS : non-significant ($p > 0.05$); S : significant ($p < 0.05$)

Region	Correlation Coefficient (r)	p-value	Significance
Frontal	-0.686	0.0601	NS
Parietal	-0.647	0.0831	NS
Temporal	-0.713	0.0472	S
Occipital	-0.708	0.0493	S
Cerebellum	-0.674	0.0666	NS

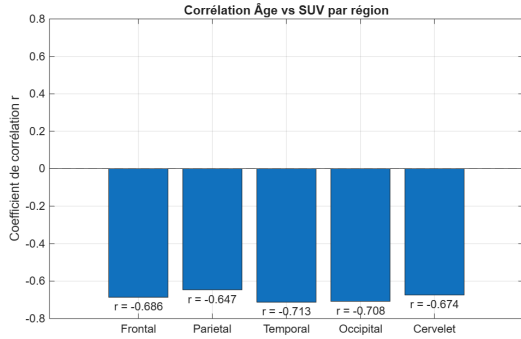


FIGURE 3. Correlation coefficients between age and regional SUVs (AD, $n=9$). Bars represent the value of r for each region. The temporal and occipital regions exhibit the strongest negative correlations ($r \approx -0.71$).

The results demonstrate moderate to strong negative correlations between age and SUVs across all studied regions (r coefficients ranging between -0.647 and -0.713). This trend indicates that as patient age advances, cerebral glucose metabolism tends to decrease.

Two regions reach statistical significance ($p < 0.05$) :

- The temporal region : $r = -0.713$ ($p = 0.0472$);
- The occipital region : $r = -0.708$ ($p = 0.0493$).

The frontal ($p = 0.0601$), parietal ($p = 0.0831$), and cerebellar ($p = 0.0666$) regions show similar trends but do not reach statistical significance, likely due to the limited sample size ($n = 8$).

These results are consistent with the pathophysiology of Alzheimer's disease : metabolic impairment is early and pronounced in temporal regions (involved in memory), whereas the relative sparing of the cerebellum (non-significant) supports its potential role as a reference region.

D. LIMITATIONS OF SUV ANALYSIS

The SUV data presented here should be interpreted with caution. Indeed, only acquisitions calibrated in Bq/mL could be converted to SUV (see Materials section). The lack of a conversion factor for CNTS data restricted our cohort to 10 patients (9 AD, 1 MCI), which limits the statistical power of intergroup comparisons. Furthermore, raw SUV values do not correct for inter-individual variations in baseline metabolism, justifying the reliance on SUVr in the following section.

E. SUVr ANALYSIS

For the SUVr calculation, we were not constrained by units because ratios are used. However, because MMSE scores were not reported for all subjects, studying this parameter proved impossible for certain patients. Consequently, our database expanded to 23 patients, including 8 CN, 6 MCI, and 9 AD. We gathered each patient's data into different tables to obtain an overall view of their profiles. This figure serves as a solid baseline to classify patients into the three study groups : CN, MCI, and AD.

TABLE 5. Clinical data by patient

ID	Group	Age	Sex	APOE	FAQ	MMSE
0295	CN	90	M	3/4	0	28
0413	CN	81	F	3/3	1	30
0685	CN	93	F	3/3	0	28
0729	MCI	69	F	3/4	10	24
1155_11	MCI	62	M	3/3	1	28
1155_13	MCI	64	M	3/3	1	30
1261	CN	77	F	3/3	0	30
1268	MCI	87	M	3/4	2	27
1280	CN	75	F	3/4	0	28
4213	CN	78	F	3/3	0	0
4225	CN	72	M	3/4	0	30
4262	CN	74	F	3/3	1	27
4799	MCI	73	M	3/3	0	0
5018	AD	73	M	4/4	17	0
0908	MCI	67	F	2/3	10	30
1059	AD	85	F	3/3	27	27
1257	AD	85	M	3/4	17	0
4136	AD	68	M	3/4	29	17
4152	AD	60	M	3/3	16	0
4373	AD	71	F	3/3	22	0
5165	AD	78	M	3/3	26	0
6264	AD	55	M	4/4	0	0
6854	AD	86	M	0	0	0

Additionally, SUVr values extracted for each patient across five key regions (frontal, temporal, parietal, occipital, and cerebellum) were gathered in a single table to compare glucose consumption across different brain areas and identify those where hypometabolism is most marked.

TABLE 6. SUVr values extracted by region of interest (ROI)

Patient ID	Frontal	Temporal	Parietal	Occipital	Cerebellum
0295	1.0259	1.0984	0.9216	1.1192	1.0000
0413	1.2117	1.3134	1.0843	1.3512	1.0000
0685	1.2276	1.2850	1.0231	1.2924	1.0000
0729	1.1826	1.2801	0.9817	1.3470	0.9999
1155_11	1.2389	1.1685	1.0428	1.1813	1.0000
1155_13	1.2153	1.1623	1.0608	1.2104	1.0000
1261	1.1731	1.2117	1.0004	1.2288	1.0000
1268	1.0162	1.0464	0.8471	1.0566	1.0000
1280	1.0647	1.2290	0.9835	1.2564	1.0000
4213	1.0780	1.1599	1.0064	1.2057	1.0000
4225	1.1846	1.2202	1.0931	1.4005	1.0000
4262	1.0735	1.1377	0.9597	1.2577	1.0000
4799	1.1246	1.1383	1.0132	1.3880	1.0000
5018	0.9418	0.9599	0.8208	1.0997	1.0000
0908	1.1192	1.1673	1.0054	1.2894	1.0000
1059	0.9457	0.9701	0.7845	1.0268	1.0000
1257	1.0716	1.1166	0.9151	1.1219	0.9999
4136	0.9418	0.8669	0.7506	0.9747	1.0000
4152	0.8390	0.8861	0.8550	1.1909	0.9999
4373	0.8845	0.9139	0.8035	1.1521	1.0000
5165	1.1043	1.0733	0.9373	1.1748	1.0000
6264	1.0722	1.0244	0.97445	1.207	1.0000
6854	0.91681	0.92682	0.78879	1.0459	1.0000

F. ANALYSIS OF THE MOST IMPACTED ZONE (SUVr)

In order to optimize the remainder of our study, we sought to base our analyses on a precise area, specifically the zone most marked by neurodegeneration. To this end, we produced the graph below, mapping metabolism by region of interest[2].

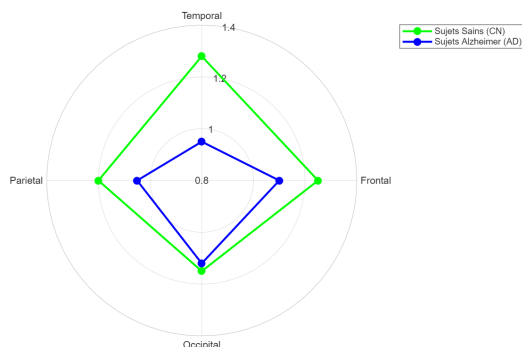


FIGURE 4. Metabolism mapping by region of interest

This plot proves that the disease does not affect the brain uniformly. The maximum divergence between healthy control subjects and affected subjects is observed along the temporal axis. While the occipital region remains spared, the temporal region drops from a mean SUVr of approximately 1.25 for a healthy subject to 0.95 for an affected subject. The temporal lobe is thus the primary focus of hypometabolism.

G. SUVr DISTRIBUTION BY DIAGNOSTIC STAGE

The figure below shows the distribution of SUVr values within the temporal region for each patient across the three study groups. From healthy control subjects to affected subjects, a clear degradation of metabolism is observed.

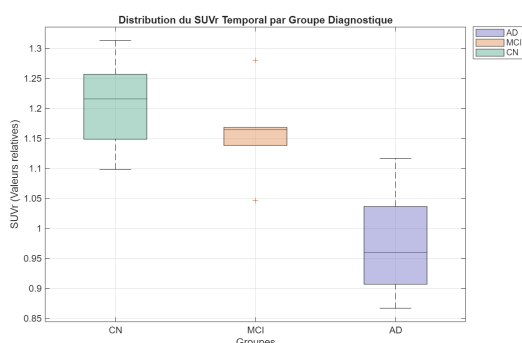


FIGURE 5. Distribution of temporal SUVr by diagnostic group

For the control group, the median SUVr is 1.22. In MCI subjects, it drops to 1.16. The majority of patients in the transition phase appear to share a nearly identical metabolism, which translates to the very compact rectangle observed in the figure. Finally, the median of the AD group lies below 1.0, positioned at approximately 0.96.

We can notice two atypical values for patients in the transition phase. One of them displays a metabolism of 1.28, consuming as much glucose as a healthy patient. Another presents a metabolism of 1.05, whose temporal region metabolism resembles that of an affected patient. Thus, other factors could explain their classification.

H. IMPACT OF METABOLISM ON THE FAQ SCORE

The FAQ (Functional Activities Questionnaire) is an assessment tool used to measure a patient's independence in

daily life. While the MMSE evaluates memory or calculation, the FAQ evaluates the actual impact of the disease on social and domestic behavior. Specifically, the questionnaire addresses ten daily activities, including the ability to shop independently, meal preparation, ability to travel independently, driving, understanding a broadcast, and following current events[13].

For each activity, the patient is assigned a score ranging from 0 to 3, accumulating up to 30 when all are combined. The figure below provides a reference table for FAQ scores.

TABLE 7. FAQ Score Reference

Score	Clinical Status	Daily Impact
0 - 5	Healthy Control (CN)	Total independence in social and domestic activities.
6 - 8	Mild Cognitive Impairment (MCI)	Occasional difficulties during complex tasks (financial management, transport).
9 - 30	Alzheimer's Disease (AD)	Marked loss of autonomy and dependence for complex daily activities.

In this section, we study the correlation between SUVr and FAQ scores. The study was performed on a sample of 15 patients selected from the provided database. The sample was divided into three sub-groups of equal size : 5 MCI patients, 5 AD patients, and 5 CN subjects.

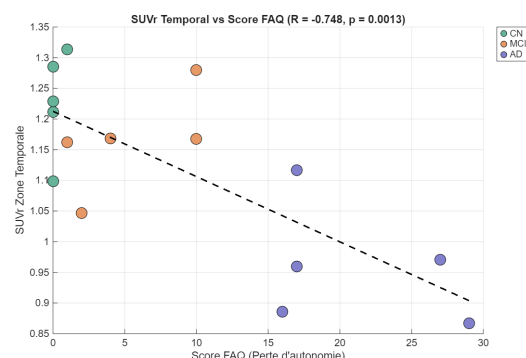


FIGURE 6. Linear correlation between functional FAQ score and cerebral metabolism (SUVr) in the temporal region.

The analysis reveals a strong negative correlation coefficient, indicating that higher FAQ scores correspond to lower SUVr values. Thus, loss of autonomy serves as a sign of neuronal death. Furthermore, the p-value of 0.0013 is well below 0.05 ; consequently, we can affirm that this relationship is statistically significant. The green points belonging to the control group cluster at the top left, exhibiting the highest SUVr values and FAQ scores near zero. The orange points belonging to the MCI group show greater dispersion, indicating a transition phase where a drop in metabolism is observed. As for the blue points belonging to the AD group, they are positioned at the bottom right, marking an advanced stage of the disease with increasingly lower SUVr values and very high FAQ scores, denoting near-total dependence.

However, an essential element warrants attention : we notice an orange point (MCI) with an FAQ score of 10 that nonetheless exhibits a very high SUVr. Consequently, the patient shows advanced signs of autonomy loss while their metabolism appears preserved. This concerns patient 0908, who also exhibited a high SUV value earlier. This profile demonstrates the importance of combining PET imaging with clinical testing.

IV. DISCUSSION

Our study successfully implemented a complete [18F]FDG PET quantification pipeline and recovered the expected metabolic signatures of Alzheimer's disease : temporal hypometabolism, occipital and cerebellar sparing, and correlation between regional metabolism and functional autonomy. Several limitations must nevertheless be highlighted.

The small size of our cohort (24 subjects, including only 10 for SUV analysis) limits statistical robustness and warrants validation on a larger sample, including longitudinal follow-up to distinguish stable MCI from converting MCI.

Five cortical regions were analyzed, but the absence of the posterior cingulate cortex, precuneus, and hippocampus (early-affected regions) might explain the difficulty in discriminating MCI patients. Moreover, the lack of age matching between groups (CN subjects being older than AD patients) does not allow us to rule out an age effect on the observed metabolic differences, as an age-SUV correlation was present in our cohort ($r = -0.71$ in the temporal lobe).

Finally, the heterogeneity of PET systems and acquisition units restricted the SUV analysis solely to patients calibrated in Bq/mL. Obtaining calibration factors and applying an harmonization protocol (EARL) would represent essential improvements to replicate this work[14].

Conclusion : Despite these limitations, our work demonstrates the feasibility of a complete PET quantification pipeline, from raw data to regional metabolic biomarkers. Temporal hypometabolism and its correlation with loss of autonomy confirm the clinical value of this approach. However, PET system heterogeneity and the lack of calibration for CNTS units remain major hurdles to generalizable quantitative application. Access to calibration factors and multicenter standardization are required prior to any large-scale clinical implementation.

GLOSSARY

AD/MA : Alzheimer's Disease / Maladie d'Alzheimer
Progressive neurodegenerative disorder.

CN : Cognitively Normal
Subjects without cognitive decline, used as metabolic references in the study.

CNTS : Raw counts
Raw, uncalibrated PET acquisition units.

FWHM : Full Width at Half Maximum
Parameter used for Gaussian smoothing of PET images.

MCI : Mild Cognitive Impairment
Transitional stage between normal aging and dementia.

MMSE : Mini-Mental State Examination
Clinical test evaluating cognitive functions (memory, calculation, language). Score out of 30.

SPM : Statistical Parametric Mapping
MATLAB toolbox used for PET image preprocessing.

SUV : Standardized Uptake Value
Quantitative index normalizing PET activity by patient weight and injected dose. Unitless.

SUVr : Standardized Uptake Value ratio
Ratio between target region activity and reference region activity (here, the cerebellum).

TEP/PET : Positron Emission Tomography / Tomographie par émission de positons
Functional imaging technique measuring metabolic activity via a radioactive tracer.

[18F]FDG : Fluorine-18 labeled Fluorodeoxyglucose
Glucose analogue used as a PET tracer to evaluate cerebral glucose consumption.

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